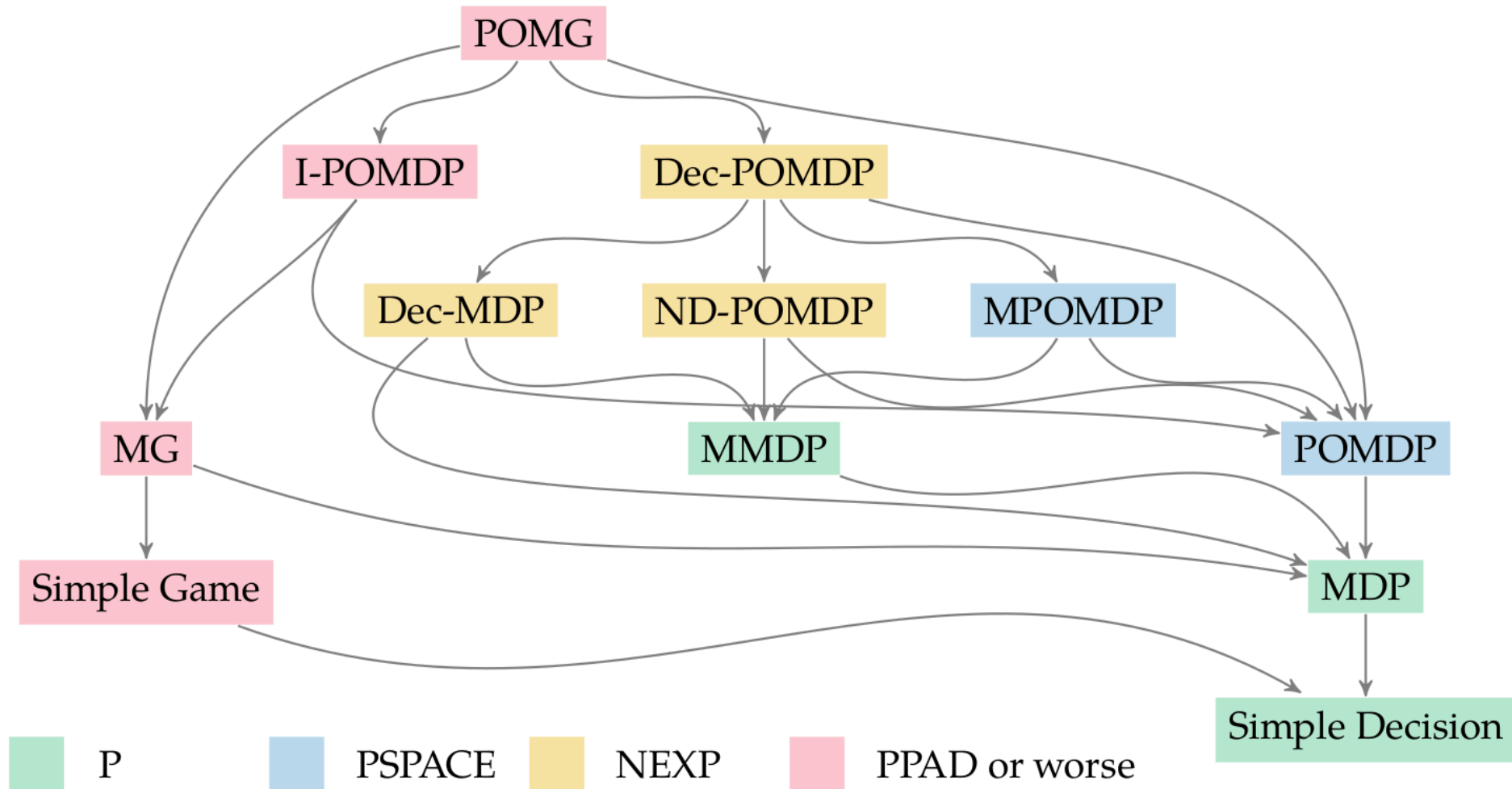


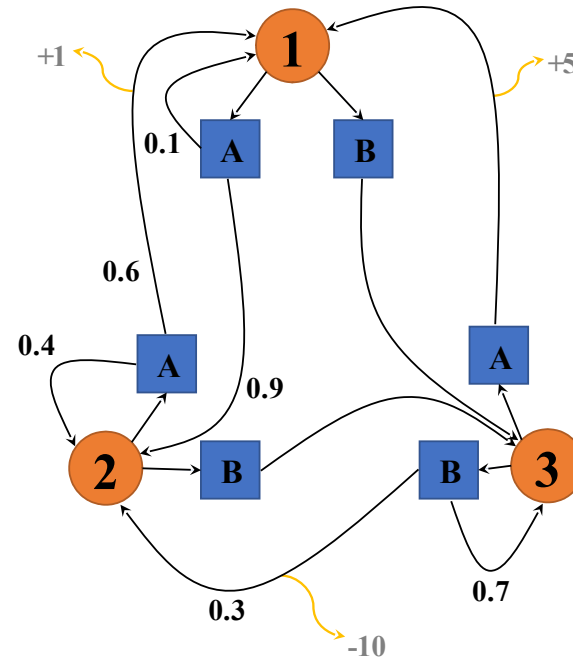
Imperfect Information Dynamic Games

Tyler Becker

Hierarchy of Problems



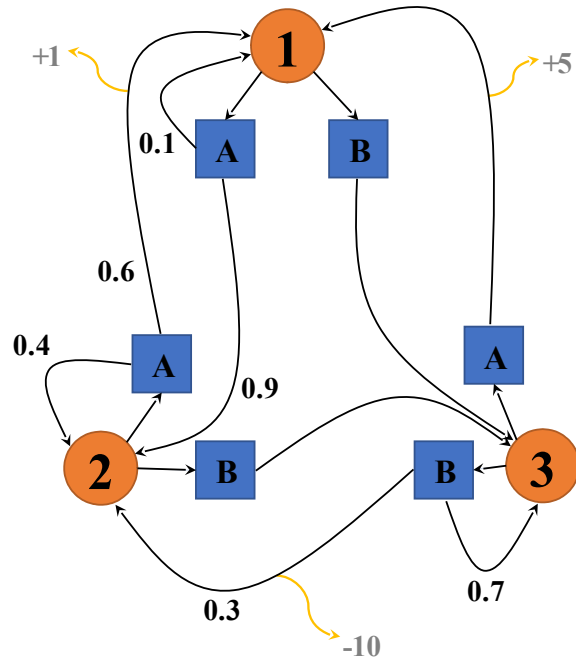
Markov Decision Process (MDP)



- $\mathcal{S}$ - State space
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ - Reward

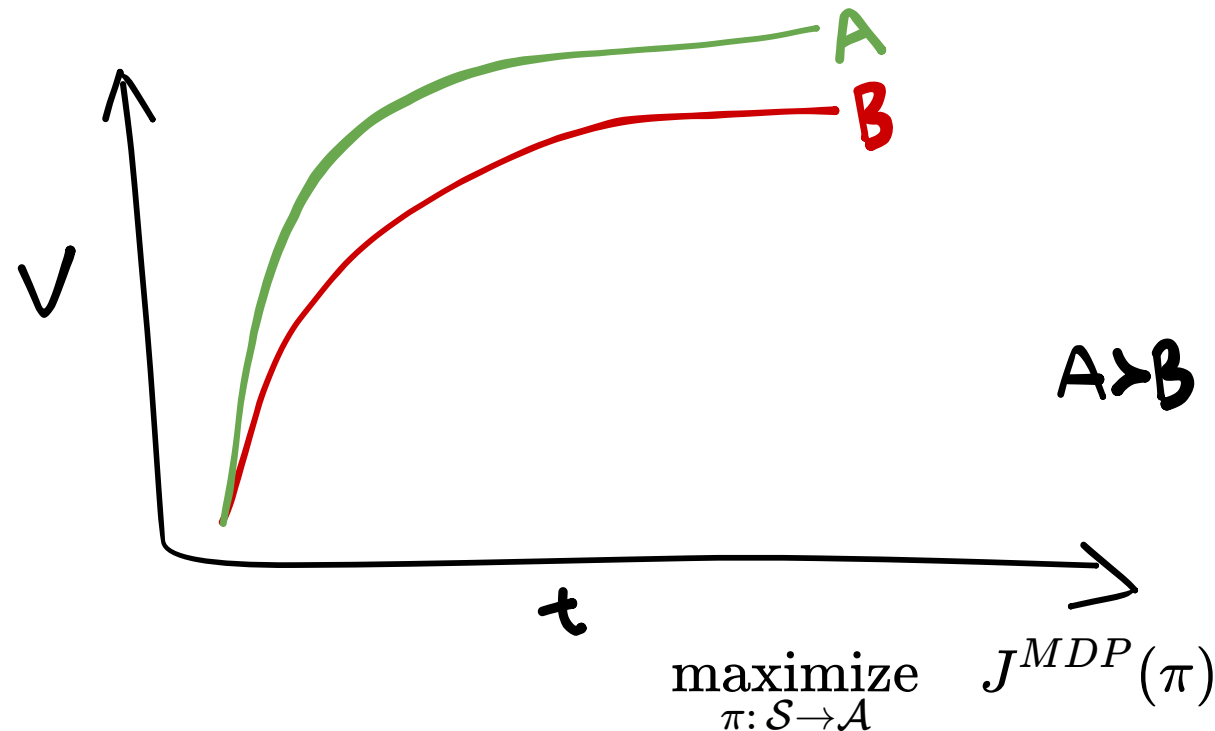
Aleatory

Markov Decision Process (MDP)

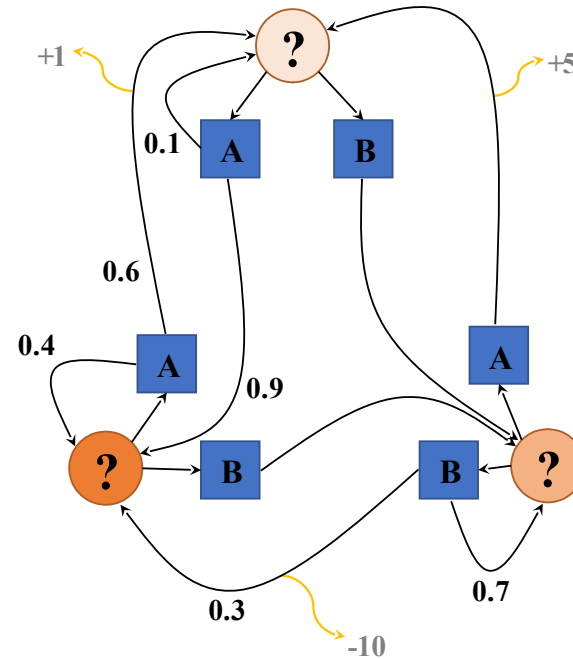


$$J^{MDP}(\pi) = V^{MDP}(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \middle| a_t = \pi(s_t) \right]$$

How can we judge a solution as better or worse?



Partially Observable Markov Decision Process (POMDP)

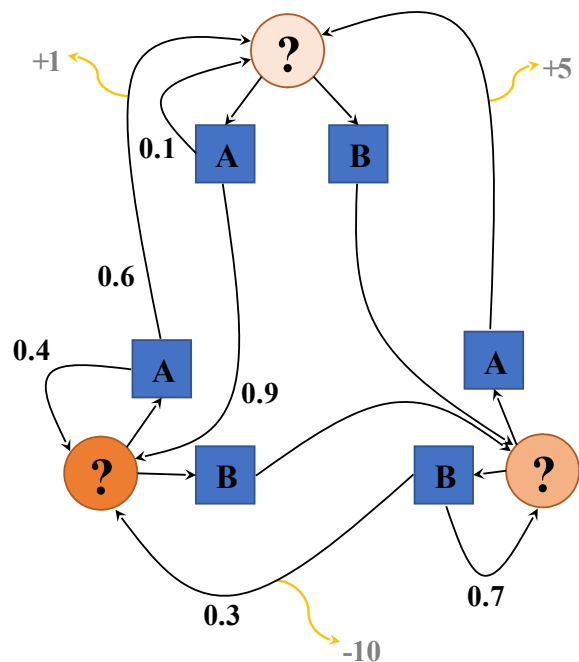


- $\mathcal{S}$ - State space
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ - Transition probability distribution
- $\mathcal{A}$ - Action space
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ - Reward
- $\mathcal{O}$ - Observation space
- $Z : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times \mathcal{O} \rightarrow \mathbb{R}$ - Observation probability distribution

Aleatory

Epistemic

Partially Observable Markov Decision Process (POMDP)



$$h_t = (a_0, o_0, \dots, a_t, o_t)$$

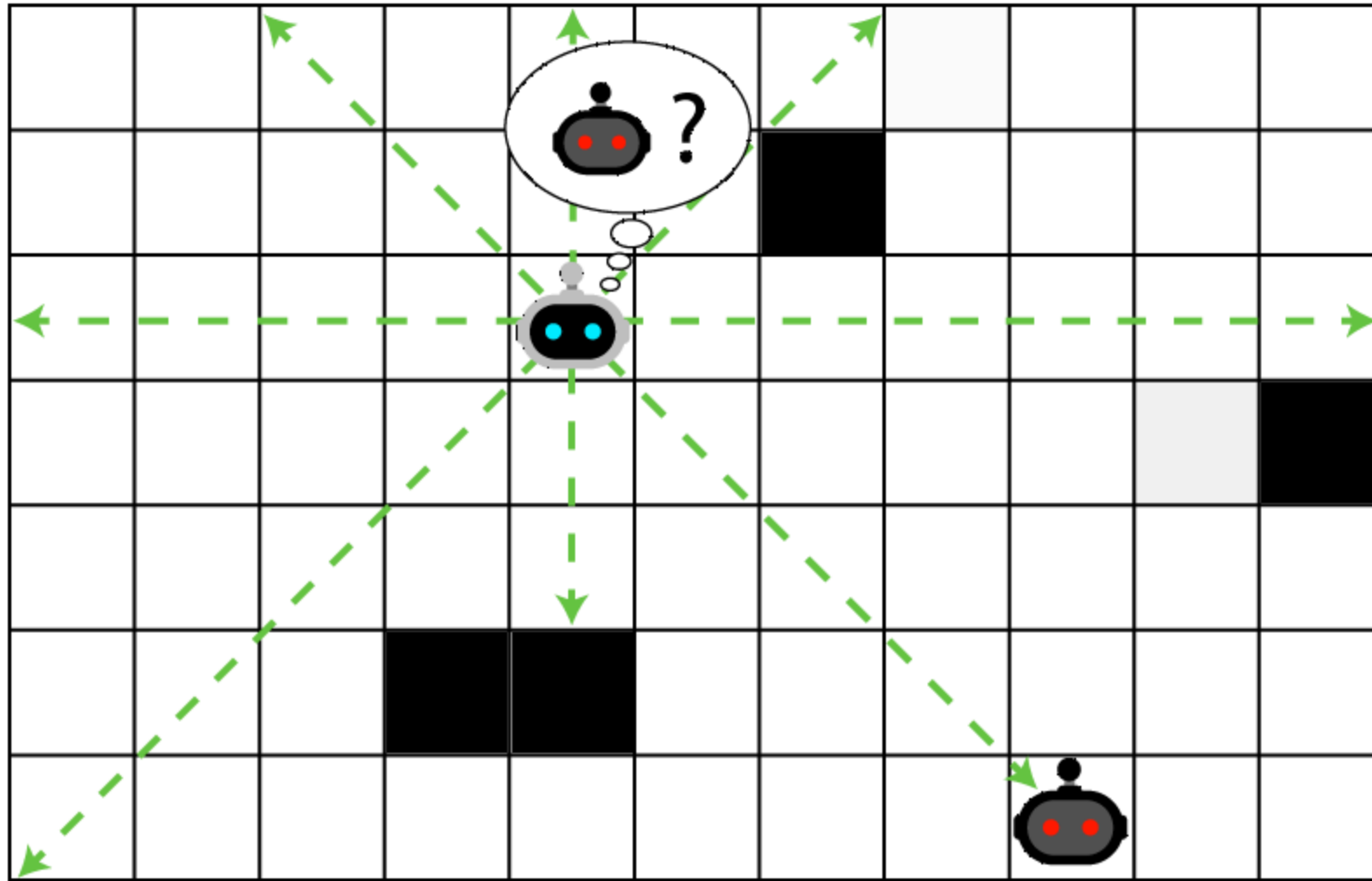
$$J^{POMDP}(\pi) = V^{POMDP}(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid a_t = \pi(\overset{b_t(s_t)}{h_t}) \right]$$

$$b_t(s_t) = P(s_t | h_t)$$

$$\underset{\pi: \mathcal{B} \rightarrow \mathcal{A}}{\text{maximize}} \quad J^{POMDP}(\pi)$$

$b_t(s_t)$

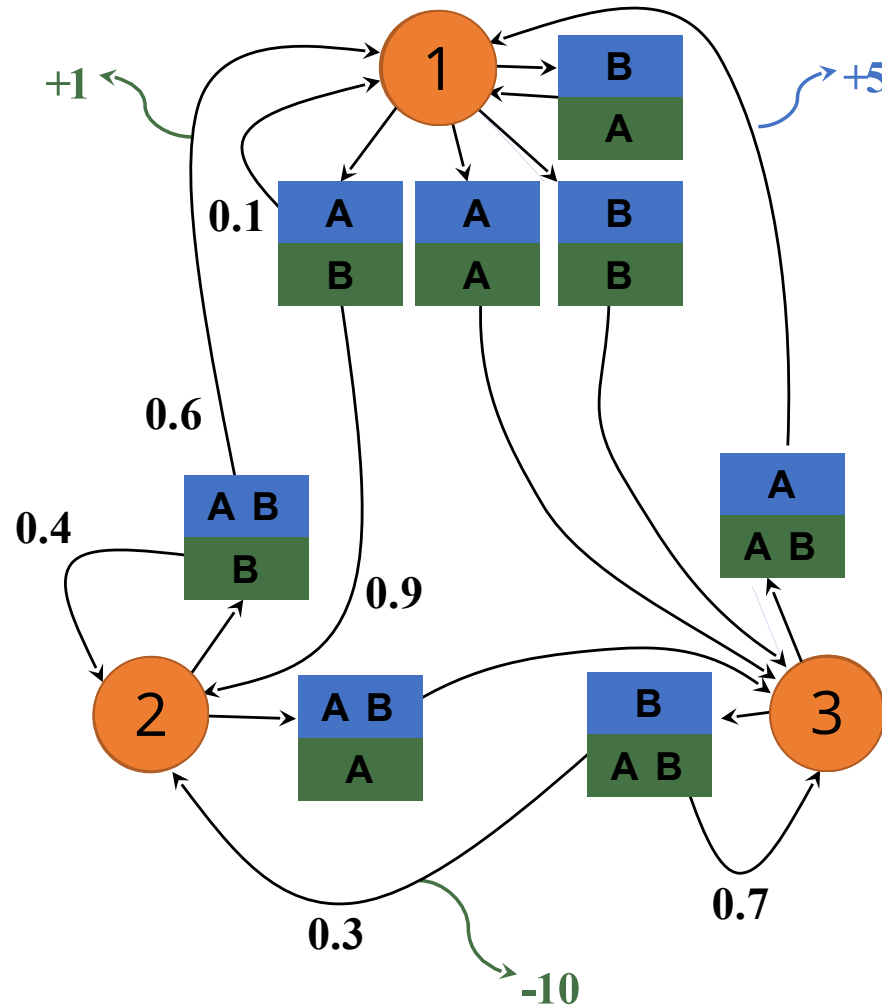
Laser Tag POMDP



Multi-agent environment posed as a single-agent problem

Evader policy predictably cowers in a corner

Markov Game (MG)

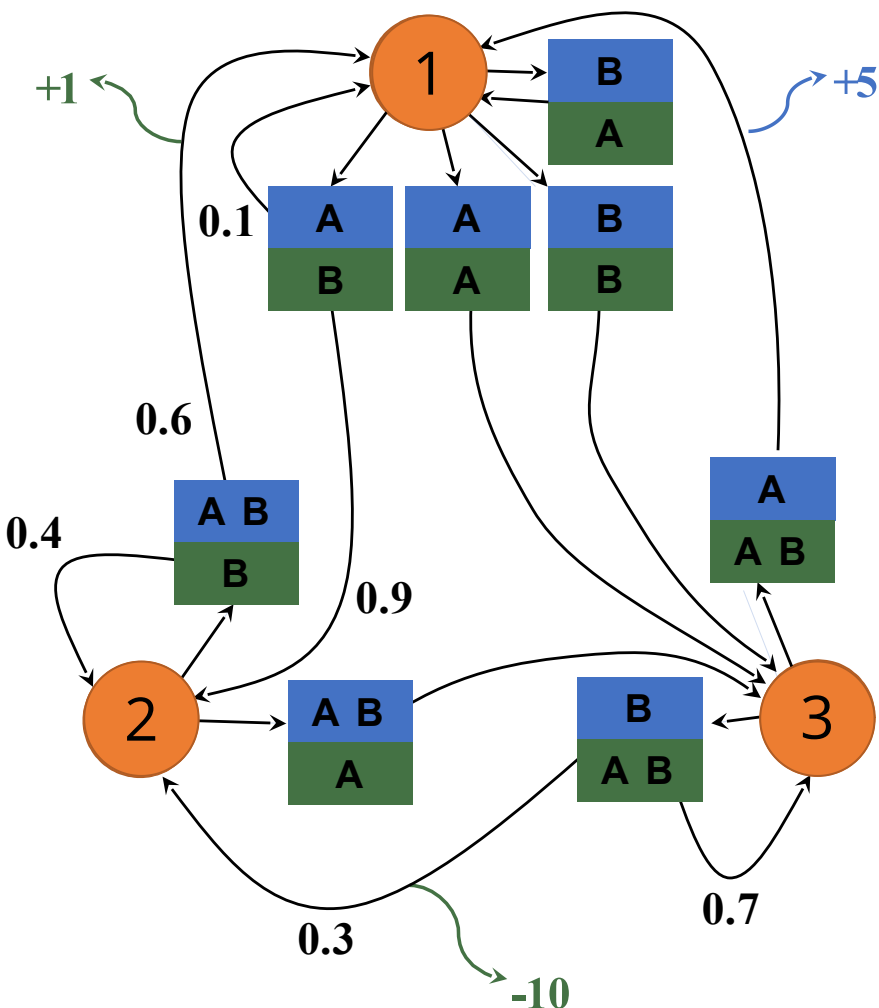


- $\mathcal{S}$ - State space
- $T(s' | s, \mathbf{a})$ - Transition probability distribution
- $\mathcal{A}^i, i \in 1..k$ - Action spaces
- $R^i(s, \mathbf{a})$ - Reward function (cooperative, opposing, or somewhere in between)

Aleatory

Interaction

Markov Game (MG)



$$V^1(\pi^1, \pi^2) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R^1(s_t, a_t^1, a_t^2) \mid a_t^1 \sim \pi^1(s_t), a_t^2 \sim \pi^2(s_t) \right]$$

If we're playing purely cooperatively, then $R^1 = R^2$ and we can take

$$J(\pi) = V(\pi) = V^1(\pi) = V^2(\pi)$$

What about noncooperative games?

For now you've been operating on the draconic rules of NASH or NOT NASH.

Agents have NO incentive to unilaterally deviate.

$$V(\pi^{i'}, \pi^{-i}) - V(\pi^i, \pi^{-i}) \leq 0 \quad \forall \pi^{i'}$$

Nash Equilibrium

Incentive to deviate makes a policy suboptimal

For a single agent:

$$\delta(\pi) = \max_{\pi'} V(\pi') - V(\pi)$$

$$\delta(\pi) = 0 \implies V(\pi) = \max_{\pi'} V(\pi') = V^*$$

For multiple agents:

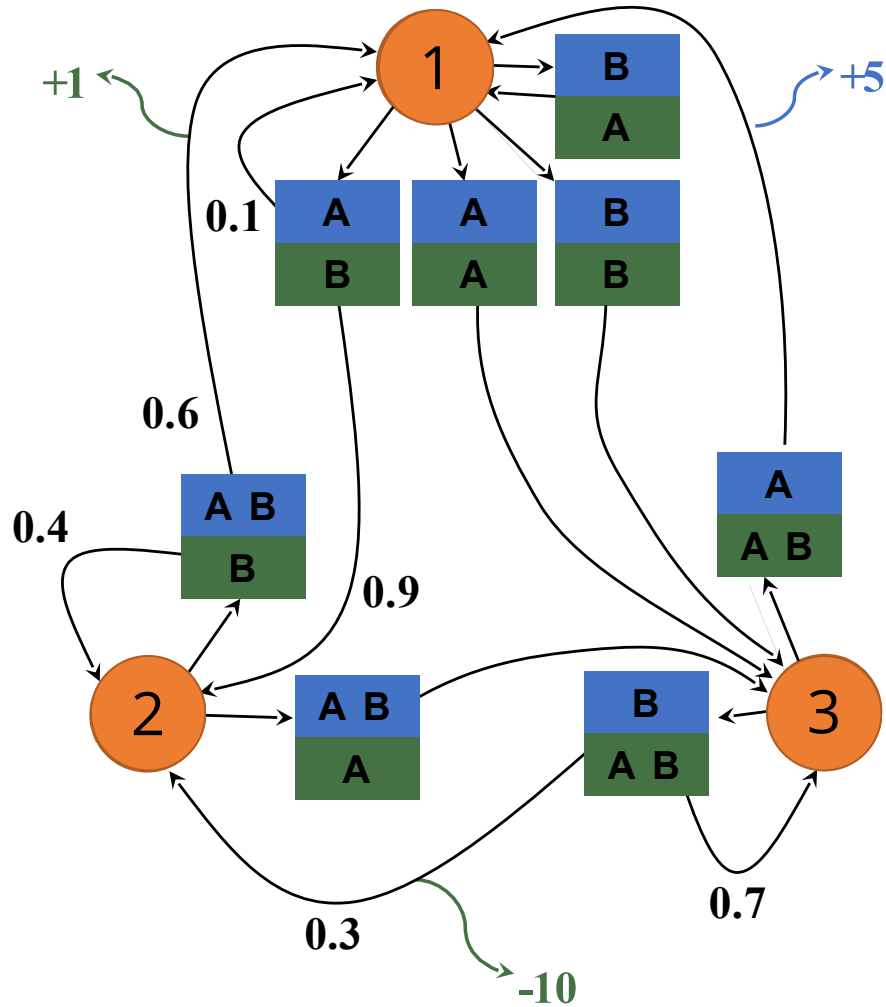
$$\delta^i(\pi^i, \pi^{-i}) = \max_{\pi^{i'}} V^i(\pi^{i'}, \pi^{-i}) - V^i(\pi^i, \pi^{-i})$$

$$\forall i, \delta^i(\pi) = 0 \implies \pi \text{ is a Nash equilibrium strategy}$$

$$\text{NashConv}_A(\pi) = \sum_i \delta_A^i(\pi), \quad \epsilon(\pi) = \max_i \delta_A^i(\pi)$$

$$V(\pi^{i'}, \pi^{-i}) - V(\pi^i, \pi^{-i}) \leq \epsilon \quad \forall \pi^{i'}$$

Markov Game (MG)



$$J(\pi) = \text{NashConv}(\pi) = \sum_i \delta^i(\pi)$$

$$\delta^i(\pi^i, \pi^{-i}) = \max_{\pi^{i'}} V^i(\pi^{i'}, \pi^{-i}) - V^i(\pi^i, \pi^{-i})$$

$$J^* = \min_{\pi} J(\pi)$$

The Prisoner's Dilemma

		Player 2	
Player 1		Cooperate	Defect
	Cooperate	<u>-1</u> , <u>-1</u>	<u>-10</u> , <u>0</u>
	Defect	<u>0</u> , <u>-10</u>	<u>-5</u> , <u>-5</u>

Two criminals are arrested for a crime. The police don't have enough evidence to convict them of a major crime, but they can convict them of a minor one. The police offer each prisoner a deal:

- If **one testifies (defects) against the other**, while the other remains silent (cooperates), the defector goes free, and the cooperator gets the full sentence (10 years).
- If **both defect**, they both get a medium sentence (5 years).
- If **both cooperate** (remain silent), they both get a short sentence (1 year).

The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	-1, -1	-10, 0
	Defect	0, -10	-5, -5

$$\pi^1 = \text{defect}, \pi^2 = \text{cooperate}$$

$$\delta^1(\pi) = 0, \quad \delta^2(\pi) = -5 - (-10) = 5$$

$$\text{NashConv}(\pi) = \sum_i \delta^i(\pi) = 0 + 5 = 5$$

The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	-1, -1	-10, 0
	Defect	0, -10	-5, -5

$$\pi^1 = \pi^2 = \text{cooperate}$$

$$\delta^1(\pi) = \delta^2(\pi) = 0 - (-1) = 1$$

$$\text{NashConv}_A(\pi) = \sum_i \delta_A^i(\pi) = 1 + 1 = 2$$

The Prisoner's Dilemma

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	-1, -1	-10, 0
	Defect	0, -10	-5, -5

$$\pi^A = \pi^B = \text{defect}$$

$$\delta^A(\pi) = \delta^B(\pi) = 0$$

$$\text{NashConv}_A(\pi) = \sum_i \delta_A^i(\pi) = 0 + 0 = 0$$

Two-player Zero-sum Games

$$R^i(s, a) = -R^{-i}(s, a) \implies V^i(\pi) = -V^{-i}(\pi)$$

How much can I stand to gain?

$$\delta^1(\pi^1, \pi^2) = \max_{\pi^{1'}} V^1(\pi^{1'}, \pi^2) - V^1(\pi^1, \pi^2)$$

How much can I stand to lose?

$$\delta^2(\pi^1, \pi^2) = V^1(\pi^1, \pi^2) - \min_{\pi^{2'}} V^1(\pi^1, \pi^{2'})$$

Again, for Nash, $\delta^i = 0 \forall i$

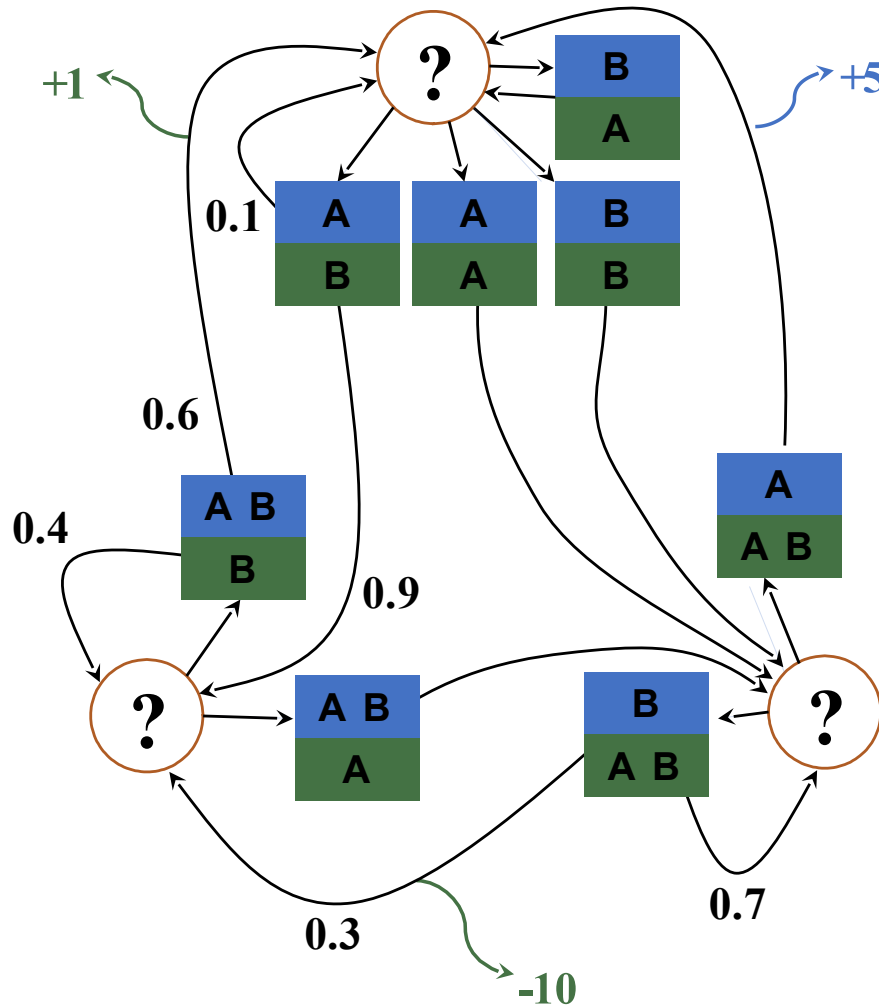
Suppose I play Nash; opponent is somewhat irrational

$$\begin{aligned} V^2(\pi^{1*}, \pi^2) &\leq V^2(\pi^{1*}, \pi^{2*}) \\ -V^2(\pi^{1*}, \pi^2) &\geq -V^2(\pi^{1*}, \pi^{2*}) \\ V^1(\pi^{1*}, \pi^2) &\geq V^1(\pi^{1*}, \pi^{2*}) \quad \forall \pi^2 \end{aligned}$$

Not only am I **guaranteed my Nash value**; opponent deviating from Nash can only increase my utility.

This is **not the case for general-sum games**. Irrational non-ego agents can bring you down with them with irrationality.

Partially Observable Stochastic Game (POSG)



- $\mathcal{S}$ - State space
- $T(s' | s, \mathbf{a})$ - Transition probability distribution
- $\mathcal{A}^i, i \in 1..k$ - Action spaces
- $R^i(s, \mathbf{a})$ - Reward function (cooperative, opposing, or somewhere in between)
- $\mathcal{O}^i, i \in 1..k$ - Observation spaces
- $Z(o^i | \mathbf{a}, s')$ - Observation probability distributions

Aleatory

Epistemic

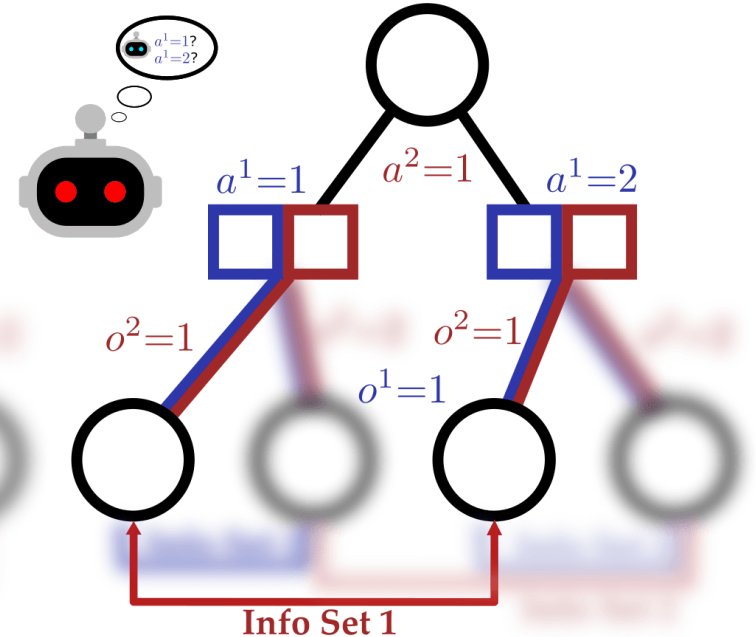
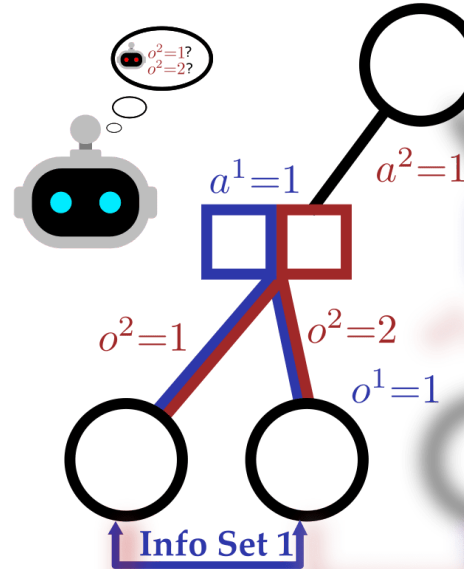
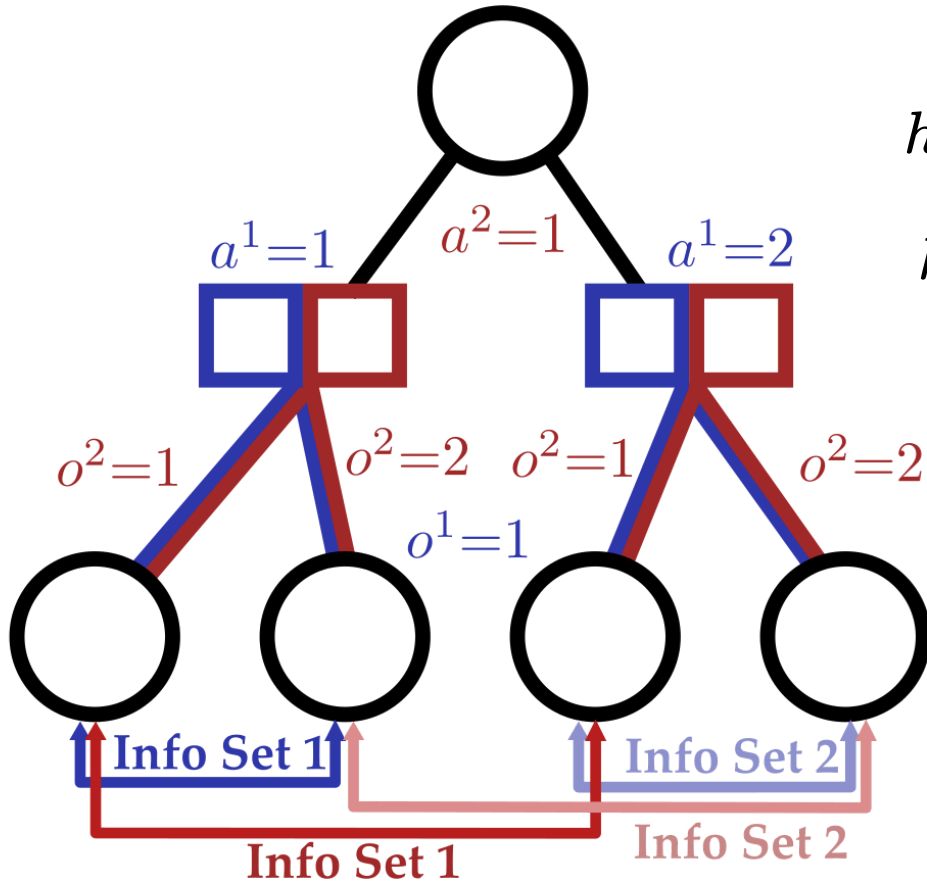
Interaction

Partially Observable Stochastic Game (POSG)

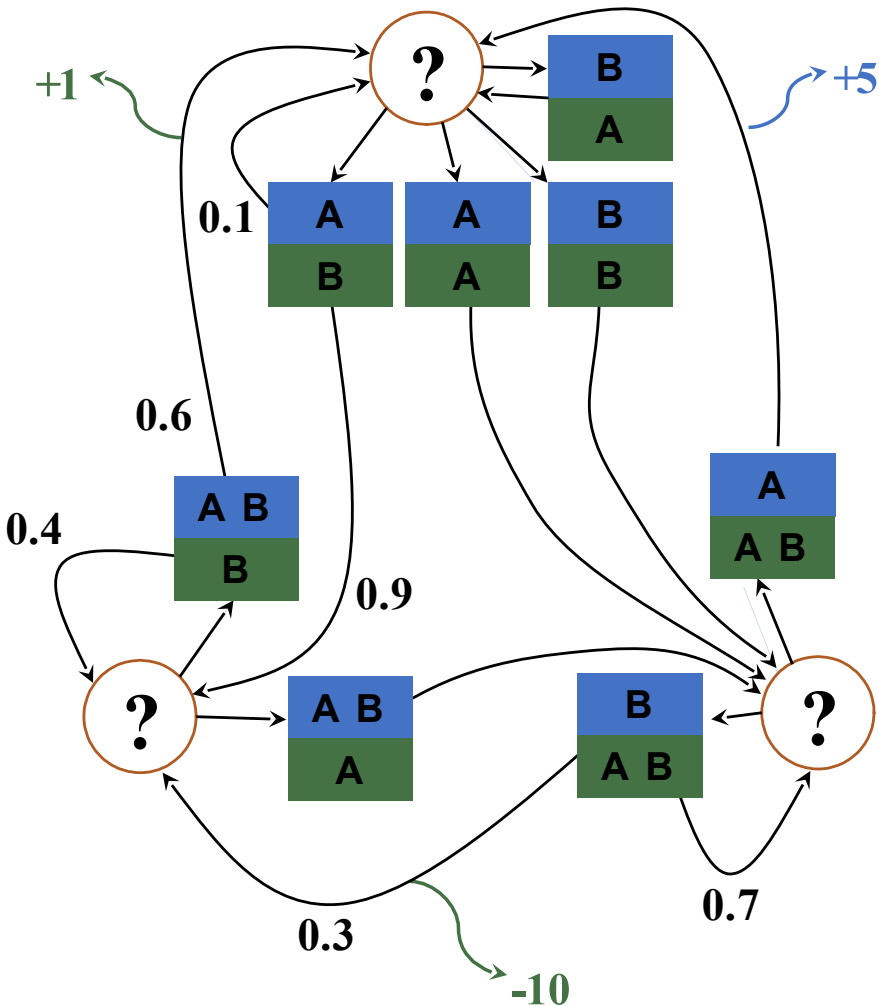
$$V^1(\pi^1, \pi^2) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R^1(s_t, a_t^1, a_t^2) \mid a_t^1 \sim \pi^1(?), a_t^2 \sim \pi^2(?) \right]$$

$$h_t = (a_0, o_0, \dots, a_t, o_t) \quad a_t = (a_t^1, a_t^2), \quad o_t = (o_t^1, o_t^2),$$

$h_t^i = (a_0^i, o_0^i, \dots, a_t^i, o_t^i)$ I only know my *own* private actions and observations



Partially Observable Stochastic Game (POSG)



$h_t^i = (a_0^i, o_0^i, \dots, a_t^i, o_t^i)$ I only know my *own* private actions and observations

What did we do with POMDPs? $a_t \sim \pi(b_t)$

$$b'(s') = P(s'|h) \propto Z(o|a, s') \sum_s T(s'|s, a) b(s)$$

$$b'(s') \propto Z(o^1, o^2 | a^1, a^2, s) \sum_s T(s'|s, a^1, a^2) b(s)$$

Problem: if we're player 1, WE DON'T KNOW a^2, o^2

- The crux of a partially observable game - we don't know what actions other agents have taken nor do we know what they've observed

Not only do we not know what state we're occupying, we don't even know what the joint history is

Policies have to be conditioned on `\emph{private histories}` (information sets/states)

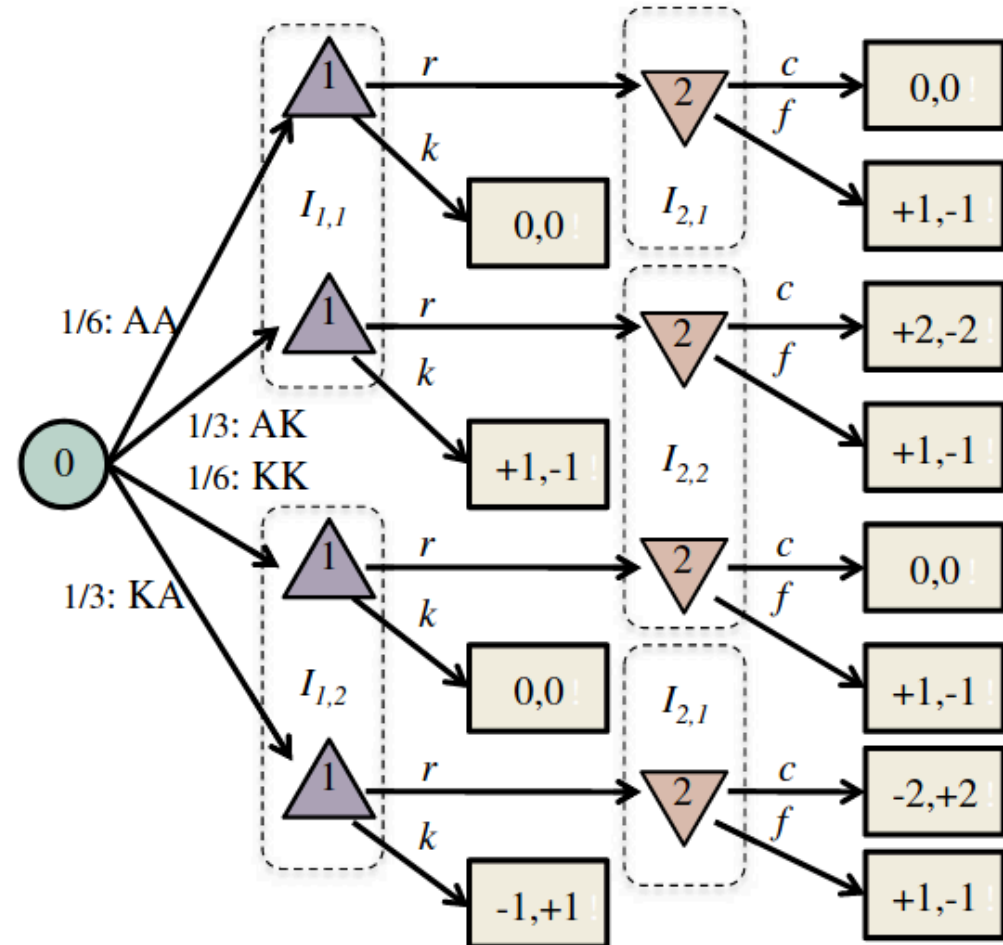
Extensive Form Game

Extensive-form game definition (h is a sequence of actions):

- Finite set of n players, plus the "chance" player
- $P(h)$ (player at each history)
- $A(h)$ (set of actions at each history)
- $I(h)$ (information set that each history maps to)
- $U(h)$ (payoff for each leaf node in the game tree)

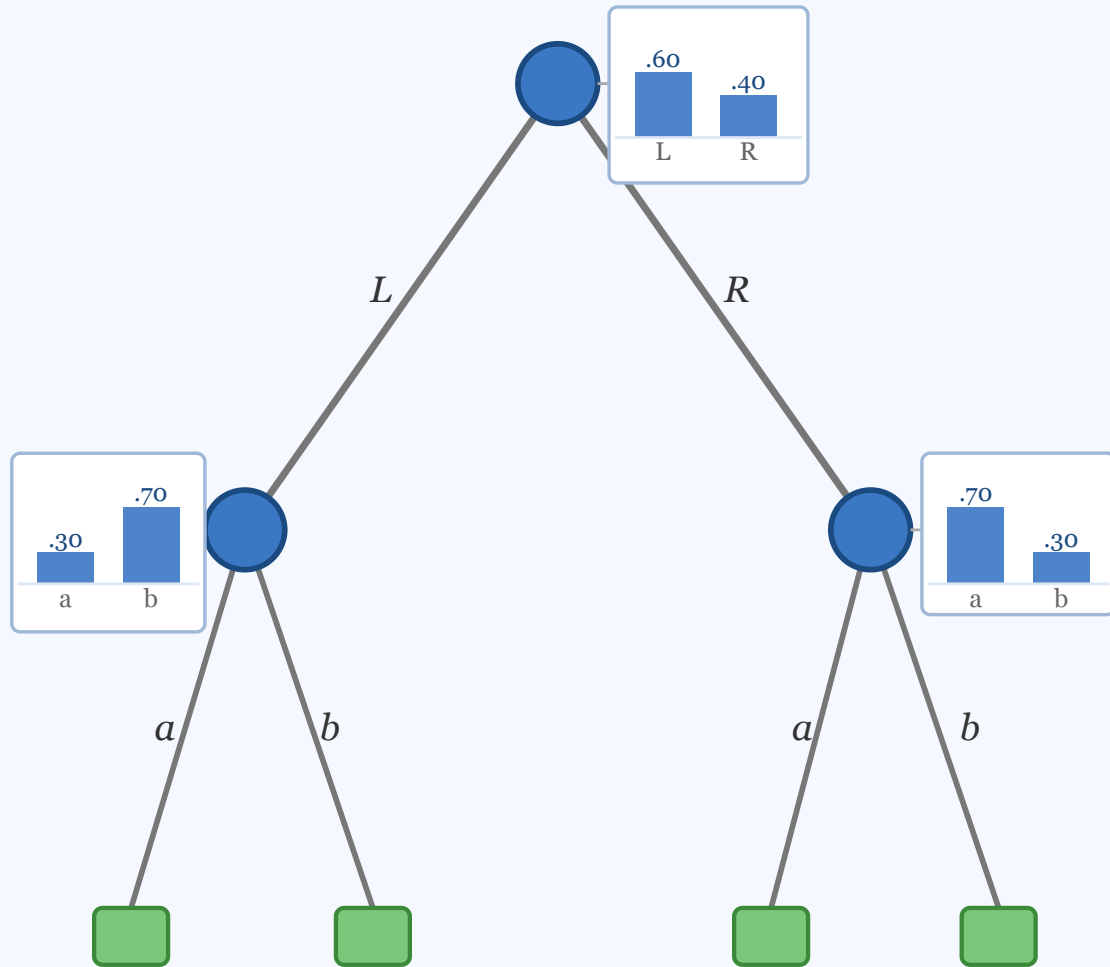
King-Ace Poker Example

- 4 Cards: 2 Aces, 2 Kings
- Each player is dealt a card
- P1 can either *raise* (r) the payoff to 2 points or *check* (k) the payoff at 1 point
- If P1 raises, P2 can either *call* (c) Player 1's bet, or *fold* (f) the payoff back to 1 point
- The highest card wins



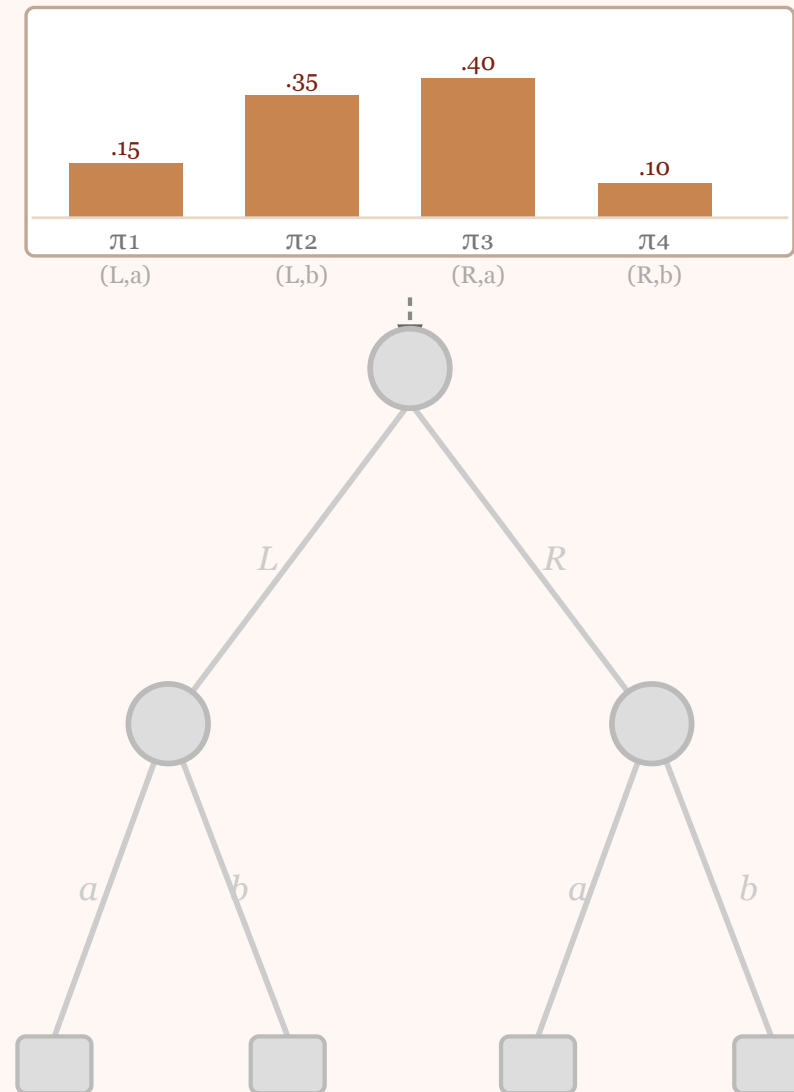
Behavioral Form

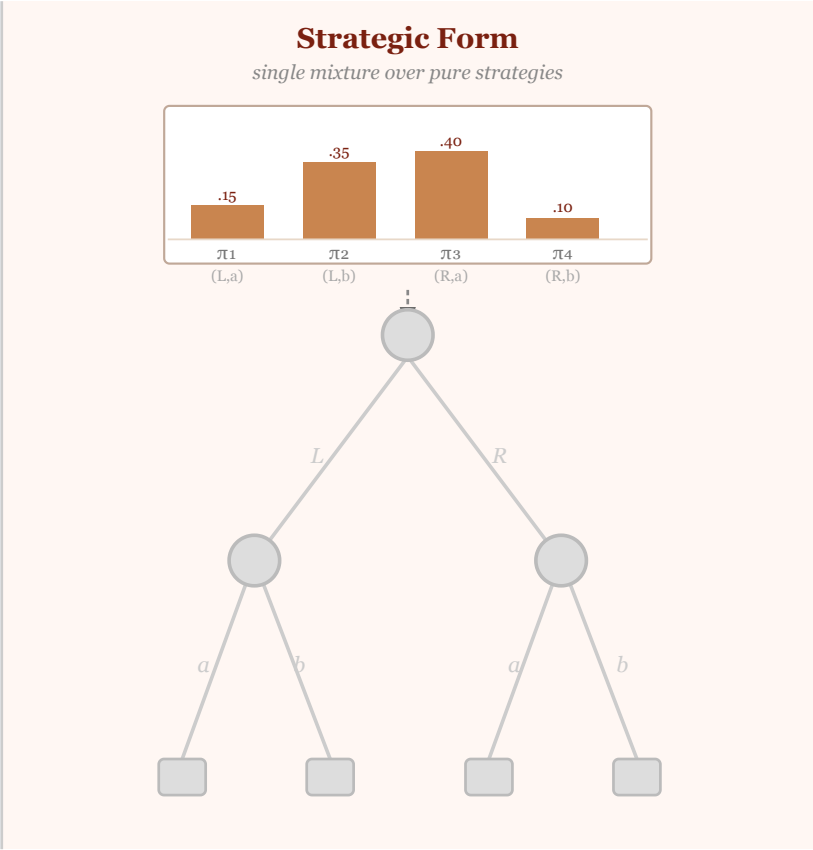
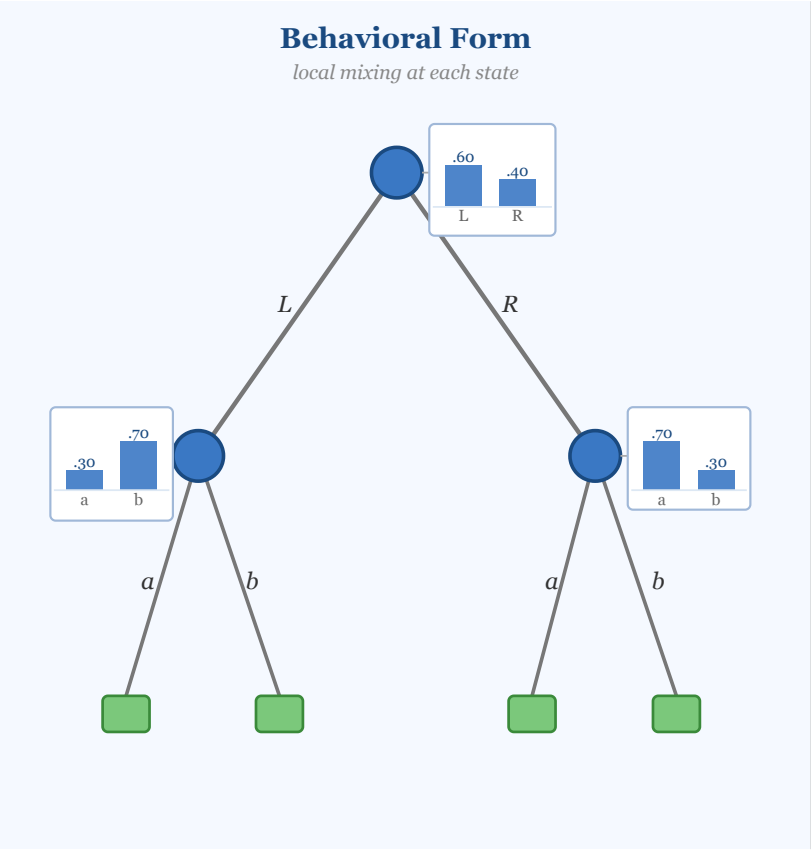
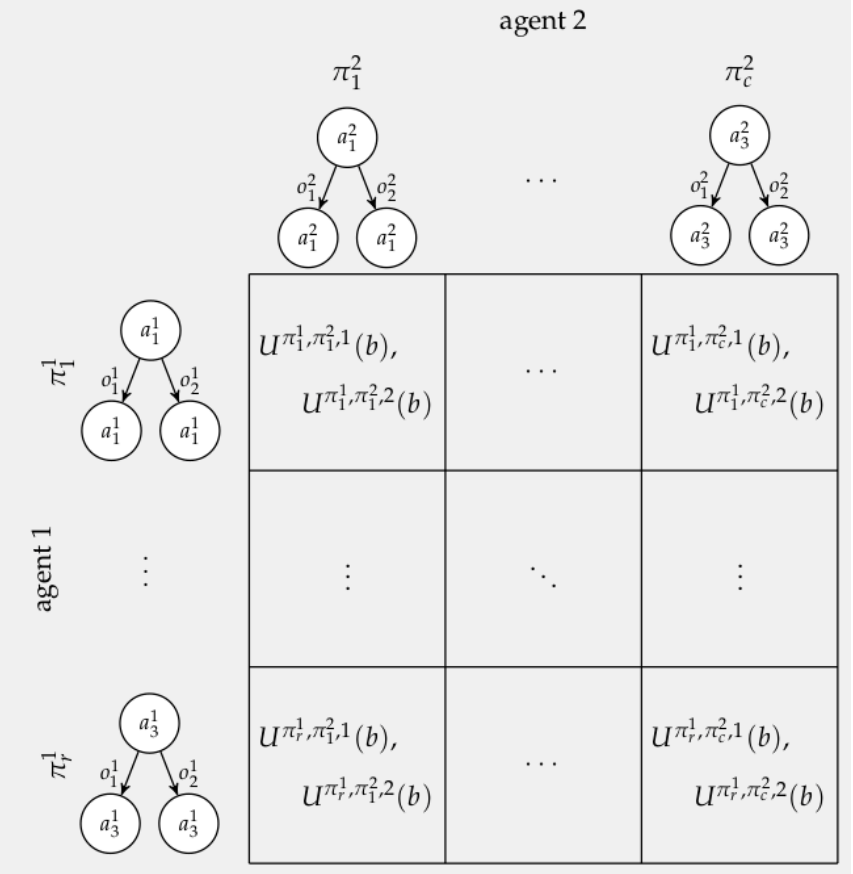
local mixing at each state



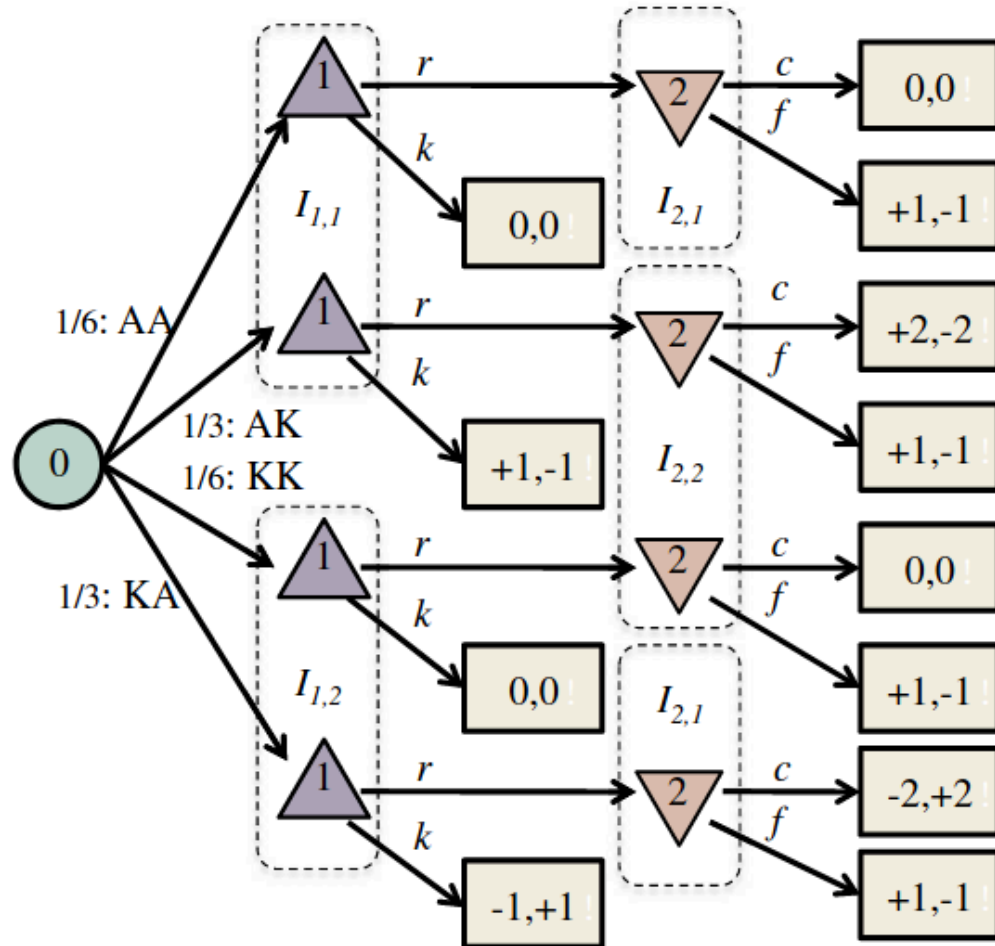
Strategic Form

single mixture over pure strategies





Extensive to Matrix Form



1:rk - raise when I have ace; check when I have king

2:cf - call when I have ace; fold when I have king

	2:cc	2:cf	2:ff	2:fc
1:rr	0	-1/6	1	7/6
1:kr	-1/3	-1/6	5/6	2/3
1:rk	1/3	0	1/6	1/2
1:kk	0	0	0	0

Exponential in number of info states!

Many methods - behavioral game theoretic reasoning is hard; strategic GT reasoning is easier

Instead of reasoning over the action space, reason over the strategy space

Fictitious Play in Extensive Form Games

Algorithm 2 General Fictitious Self-Play

function FICTITIOUSSELFPLAY(Γ, n, m)

Initialize completely mixed π_1

$\beta_2 \leftarrow \pi_1$

$j \leftarrow 2$

while within computational budget **do**

$\eta_j \leftarrow \text{MIXINGPARAMETER}(j)$

$\mathcal{D} \leftarrow \text{GENERATEDATA}(\pi_{j-1}, \beta_j, n, m, \eta_j)$

for each player $i \in \mathcal{N}$ **do**

$\mathcal{M}_{RL}^i \leftarrow \text{UPDATERLMEMORY}(\mathcal{M}_{RL}^i, \mathcal{D}^i)$

$\mathcal{M}_{SL}^i \leftarrow \text{UPDATESLMEMORY}(\mathcal{M}_{SL}^i, \mathcal{D}^i)$

$\beta_{j+1}^i \leftarrow \text{REINFORCEMENTLEARNING}(\mathcal{M}_{RL}^i)$

$\pi_j^i \leftarrow \text{SUPERVISEDLEARNING}(\mathcal{M}_{SL}^i)$

end for

$j \leftarrow j + 1$

end while

return π_{j-1}

end function

function GENERATEDATA(π, β, n, m, η)

$\sigma \leftarrow (1 - \eta)\pi + \eta\beta$

$\mathcal{D} \leftarrow n$ episodes $\{t_k\}_{1 \leq k \leq n}$, sampled from strategy profile σ

for each player $i \in \mathcal{N}$ **do**

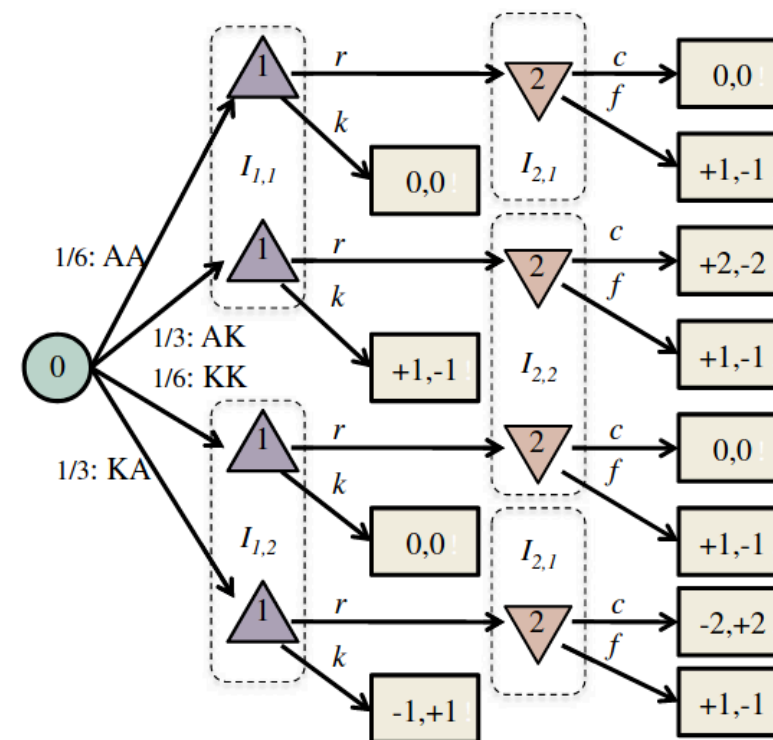
$\mathcal{D}^i \leftarrow m$ episodes $\{t_k^i\}_{1 \leq k \leq m}$, sampled from strategy profile (β^i, σ^{-i})

$\mathcal{D}^i \leftarrow \mathcal{D}^i \cup \mathcal{D}$

end for

return $\{\mathcal{D}^k\}_{1 \leq k \leq N}$

end function



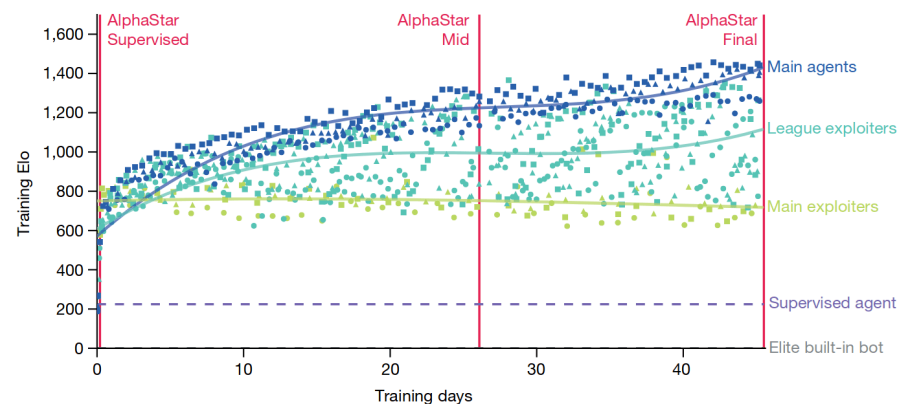
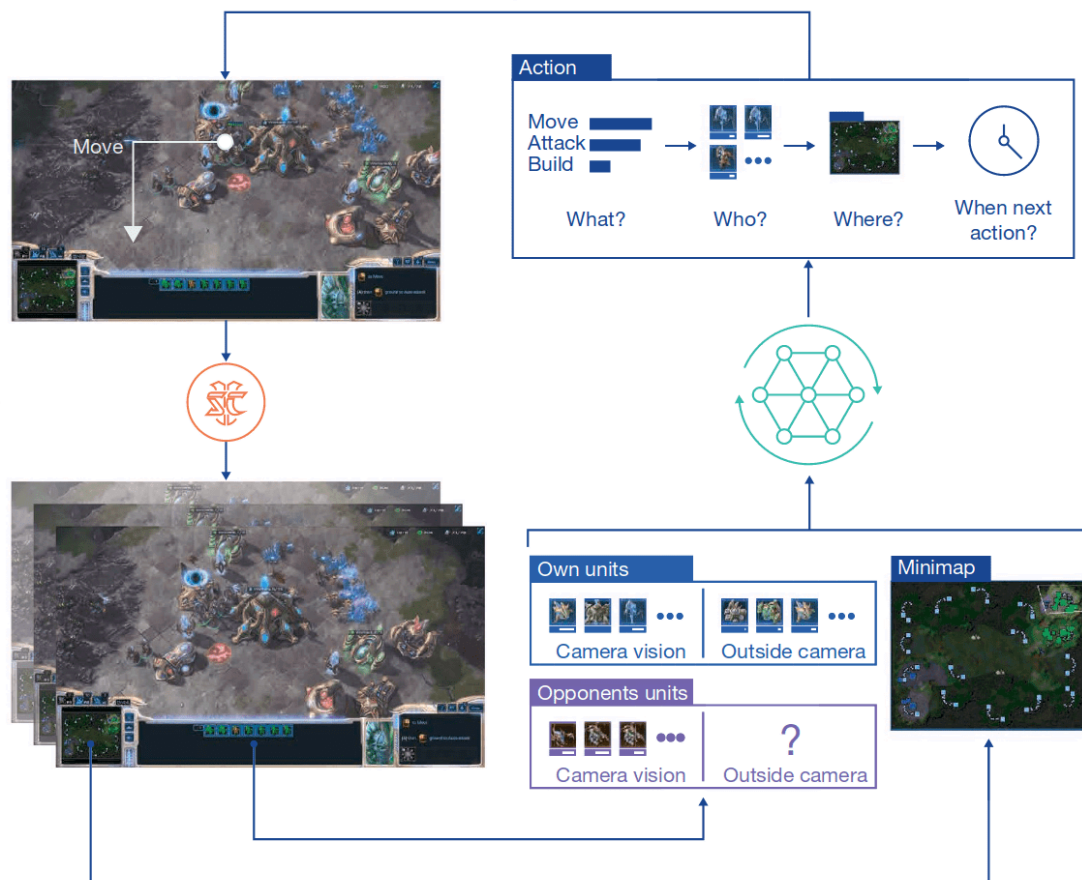
Best responses computed via RL

Mixing performed during best responses rather than with discrete jumps

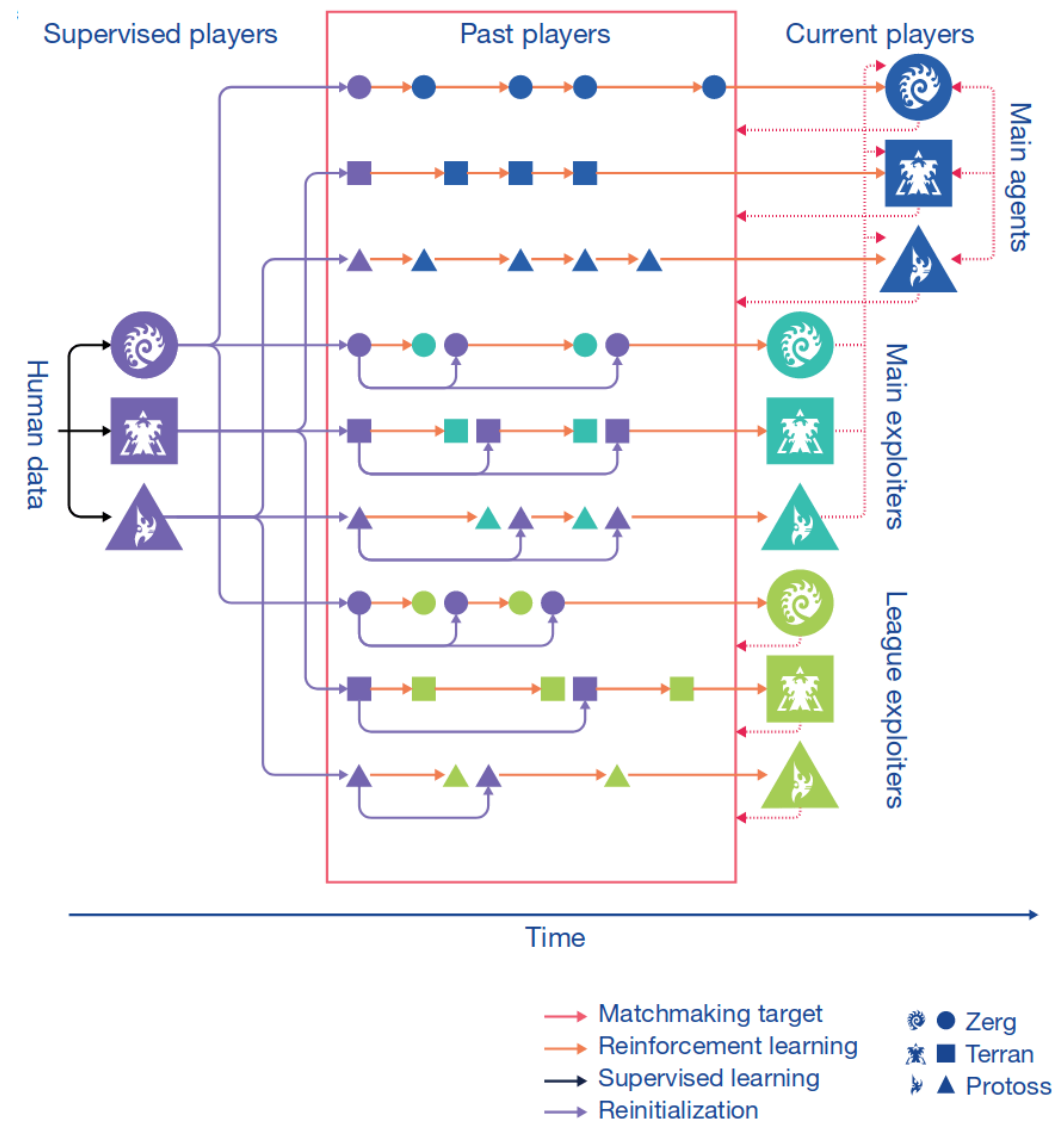
Actions limit ~22 per 5 s Requested delay ~200 ms

Real-time processing delay 80 ms

Real-time processing delay 30 ms

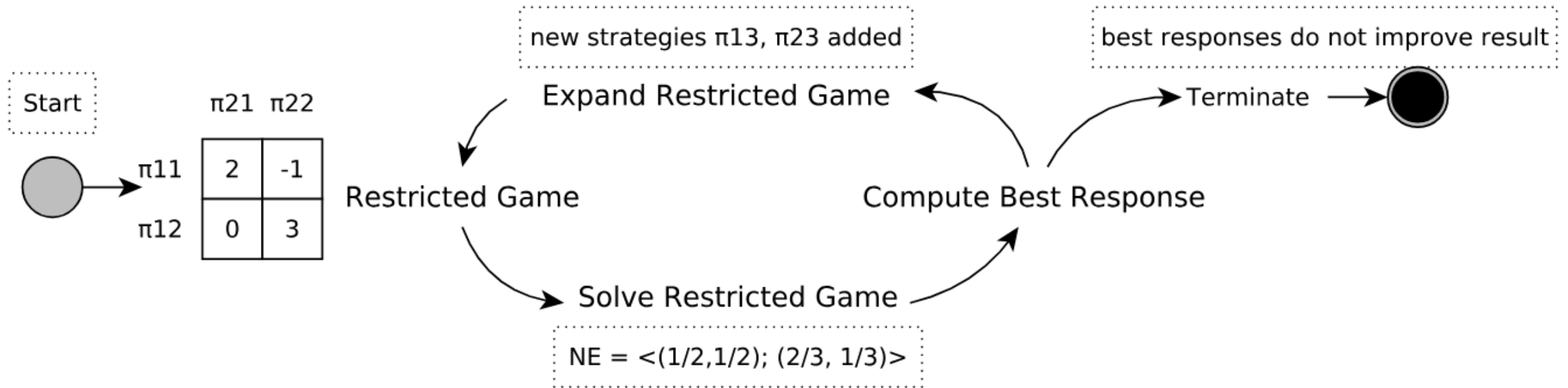


Alpha Star



Alpha Star

Seems complicated - simple premise - Policy Space Response Oracles (PSRO)
/ Double Oracle Methods



Instead of mixing uniformly over BRs, DO/PSRO explicitly solve the significantly smaller restricted strategic form game